

Multiple Choice Questions
(and solutions)
Some of the basics of each handout

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In what follows you will find a series of multiple choice questions covering some of the basics of the module. There is a set of questions for each handout. The solutions to each of the questions is given at the end of this document.

The purpose of these questions is to provide you a means to assess your understanding of many of the concepts covered and, thereby, provide you with some **formative feedback** as you go through the content.

Note that the multiple choice questions cover basics and should not be seen as a replacement of going through the concepts, derivations and problems covered in the handouts, the tutorial sheets and the recommended reading.

Handout 2

1. Consider a dynamical system

$$\dot{x} = f(x) + g(x)u,$$

and suppose you wish to design the control input u to minimise a cost functional

$$J(u, x_0) = \frac{1}{2} \int_0^\infty q(x) + u^\top R u dt,$$

where $q(x) > 0$ for all $x \neq 0$ and $q(0) = 0$, and $R = R^\top > 0$.

Which of the following statements is/are true?

- A) The term $q(x)$ penalizes the control effort, reflecting the objective to minimise fuel usage, while $u^\top R u$ ensures the state $x(t)$ is driven towards the origin.
- B) The term $\int_0^\infty q(x) dt$ penalises the final state $x(\infty)$, and the condition $q(x) > 0$ for $x \neq 0$ is needed to ensure the problem is a finite-horizon optimal control problem.
- ✓C) The function $q(x)$ represents a penalty on the state trajectory, incurring a penalty whenever $x(t) \neq 0$.
- ✓D) The term $u^\top R u$ represents a penalty on the control effort.

2. Consider a linear, discrete-time system

$$x(k+1) = Ax(k) + Bu(k).$$

Suppose we wish to design the control input u to minimise some cost functional. Which of the following cost functional/functionals result/results in an LQR problem for the system in question?

✓ A)

$$J(u, x_0) = \frac{1}{2} \sum_{k=0}^{\infty} (x(k)^{\top} Q x(k) + u(k)^{\top} R u(k)),$$

for some appropriate selections of the weighting matrices Q and R .

✓ B)

$$J(u, x_0) = \frac{1}{2} \sum_{k=0}^{N-1} (x(k)^{\top} Q x(k) + u(k)^{\top} R u(k)) + \frac{1}{2} x(N)^{\top} Q_N x(N),$$

for some appropriate selections of the weighting matrices Q , Q_N and R .

C)

$$J(u, x_0) = \frac{1}{2} \int_0^{\infty} (x(k)^{\top} Q x(k) + u(k)^{\top} R u(k)) dt,$$

for some appropriate selections of the weighting matrices Q and R .

D)

$$J(u, x_0) = \frac{1}{2} \int_0^{N-1} (x(k)^{\top} Q x(k) + u(k)^{\top} R u(k)) dt + \frac{1}{2} x(N)^{\top} Q_N x(N),$$

for some appropriate selections of the weighting matrices Q , Q_N and R .

Handout 3

1. Consider a dynamical system

$$x(k+1) = f(x(k)) + g(x(k))u(k),$$

and suppose you wish to design the control input u to minimise a cost functional

$$J(u, x_0) = \frac{1}{2} \sum_{k=0}^{N-1} q(x) + u^\top R u + q_N(x(N)),$$

where $q(x) \geq 0$ and $R = R^\top R > 0$.

Which of the following is the definition of *value function*?

- A) The average cost accumulated over the entire time horizon, normalised by the terminal time N .
- B) The instantaneous cost of the cost functional's integrand, *i.e.* $\frac{1}{2}q(x) + u^\top R u$, evaluated at the optimal control input $u^{star}(k)$ at each time $k = 0, \dots, N-1$.
- ✓ C) The minimum possible cost accumulated from time step k to the final time N , given that the system is in state x_k at time k .
- D) The minimum cost incurred up to time step $k-1$, given the initial state x_0 .

2. The Bellman equation is the cornerstone of dynamic programming, enabling a backwards recursive solution of discrete-time optimal control problems. For a discrete-time, finite-horizon optimal control problem involving a system with dynamics

$$x_{k+1} = \tilde{f}(x_k, u_k),$$

and cost functional

$$J(u, x_0) = \sum_{k=0}^{N-1} L(x_k, u_k) + \Phi(x_N),$$

which of the following is the Bellman equation?

✓ A)

$$V(x, k) = \min_u \{L(x, u) + V(\tilde{f}(x, u), k + 1)\}.$$

B)

$$V(x, k + 1) = \min_u \{L(x, u) + V(x, k)\}.$$

C)

$$V(x, k - 1) = \min_u \{L(x, u) + V(x, k + 1)\}.$$

D)

$$V(x, k) = \min_u \{L(x(k + 1), u(k + 1)) + V(x(k + 1), k + 1)\}.$$

Handout 4

1. Consider a dynamical system

$$\dot{x} = f(x) + g(x)u,$$

and suppose you wish to design the control input u to minimise a cost functional

$$J(u, x_0) = \frac{1}{2} \int_0^\infty q(x) + u^\top R u dt,$$

where $q(x) > 0$ for all $x \neq 0$ and $q(0) = 0$, and $R = R^\top > 0$.

Which of the following statements is/are true?

- ✓ A) The value function $V(x(t))$ can be defined as a function of x only – without including an explicit dependence on time due to the infinite-horizon setting and due to the problem being time-invariant.
- ✓ B) If the value function $V(x)$ associated with the problem is known, then the $V(x_0)$ denotes the best possible cost that can be achieved from the initial state $x(0) = x_0$.
- ✓ C) The HJB partial differential equation associated with this problem is

$$\frac{1}{2}q(x) + \frac{\partial V}{\partial x} f(x) - \frac{1}{2} \frac{\partial V}{\partial x} g(x) R^{-1} g(x)^\top \frac{\partial V}{\partial x}^\top = 0,$$

where $\frac{\partial V}{\partial x}$ denotes the *row* vector containing the partial derivatives of $V(x)$ with respect to x .

- D) The HJB partial differential equation associated with this problem is

$$\frac{1}{2}q(x) + \frac{\partial V}{\partial x} f(x) - \frac{1}{2} \frac{\partial V}{\partial x} g(x) R^{-1} g(x)^\top \frac{\partial V}{\partial x}^\top = 0,$$

where $\frac{\partial V}{\partial x}$ denotes the *column* vector containing the partial derivatives of $V(x)$ with respect to x .

$$H(x, u, \lambda) = \frac{1}{2}q(x) + \frac{1}{2}u^\top R u + \lambda^\top (f(x) + g(x)u)$$

$$\therefore u^* = -R^{-1} g(x)^\top \lambda$$

$$\therefore H(x, u, \lambda) = \frac{1}{2}q(x) + \frac{1}{2}\lambda^\top g(x) R^{-1} g(x)^\top \lambda + \lambda^\top f(x) + \lambda^\top g(x) R^{-1} g(x)^\top \lambda$$

$$= \frac{1}{2}q(x) + \lambda^\top f(x) - \frac{1}{2}\lambda^\top g(x) R^{-1} g(x)^\top \lambda$$

Handout 5

1. Consider the LQR problem defined by a system with dynamics

$$x_{k+1} = Ax_k + Bu_k,$$

and the cost functional

$$J(u, x_0) = \frac{1}{2} \sum_{k=0}^{N-1} (x_k^T Q x_k + u_k^T R u_k) + \frac{1}{2} x_N^T Q_N x_N,$$

where $Q = Q^T > 0$, $Q_N = Q_N^T > 0$, and $R = R^T > 0$. In this setting the optimal control can be found via one of the following equations – which one? Note that in all of the below it is assumed that $P = P^T \geq 0$

A)

$$P = Q + A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A$$

✓ B)

$$P_k = A^T P_{k+1} A - A^T P_{k+1} B (R + B^T P_{k+1} B)^{-1} B^T P_{k+1} A + Q, \quad P_N = Q_N$$

C)

$$A^T P + P A - P B R^{-1} B^T P + Q = 0$$

D)

$$-\dot{P}(t) = P(t) A + A^T P(t) - P(t) B R^{-1} B^T P(t) + Q, \quad P(N) = Q_N$$

2. Consider the LQR problem defined by dynamics

$$\dot{x} = Ax + Bu,$$

and cost functional

$$J(u, x_0) = \frac{1}{2} \int_0^\infty x^\top Q x + u^\top R u dt.$$

Suppose that $Q = Q^\top > 0$, $R = R^\top > 0$, and that $P = P^\top > 0$ is a solution of

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0.$$

Which of the following statements is/are true?

- ✓ A) The best possible cost that can be achieved from $x(0) = x_0$ is $V(x_0) = \frac{1}{2}x_0^\top P x_0$.
- ✓ B) $V(x) = \frac{1}{2}x^\top P x$ satisfies the HJB partial differential equation associated with this optimal control problem.
- ✓ C) The system in closed-loop with $u = -R^{-1}B^\top P x$ is stable.
- ✓ D) The optimal control is a linear feedback strategy with the optimal gain given by $K^* = -R^{-1}B^\top P$.

Handout 6

1. Willem's Fundamental Lemma states that if collected input-state data satisfies a persistency of excitation condition, any trajectory of a linear system can be represented by the collected data. Using the notation introduced in Handout 6 and considering a linear discrete-time system with n states and m control inputs, what must be the rank of the data matrix $\begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix}$ for the data to be persistently exciting and what does this imply about the minimum duration (T) of the experiment/simulation used to generate the data?
 - A) The rank must be equal to $n + m$, and T must be greater than $n \cdot m$.
 - B) The rank must be equal to T ; T must be greater than or equal to $n \cdot m$.
 - ✓ C) The rank must be equal to $n + m$; T must be greater than or equal to $n + m$.
 - D) The rank must be greater than $n + m$; T must be greater than $n + m$.
2. A key result of the data-driven approach considered in Handout 6 is the ability to represent the closed-loop system dynamics, $x(k+1) = (A+BK)x(k)$, purely in terms of data, without identifying A and B . This representation is given by $x(k+1) = X_{1,T}G_Kx(k)$, where G_K is a matrix that must satisfy which constraint, linking it to the desired feedback gain K and the collected data?

A)

$$X_{0,T-1}G_K = X_{1,T}$$

B)

$$X_{1,T}G_K = I$$

✓ C)

$$\begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix} G_K = \begin{bmatrix} I \\ K \end{bmatrix}$$

D)

$$U_{0,T-1}G_K = K$$

Handout 7

1. Consider a dynamical system

$$\dot{x} = f(x) + g(x)u,$$

with $x \in \mathbb{R}^n$ and $u \in \mathbb{R}^m$, and suppose you wish to design the control input u to minimise a cost functional

$$J(u, x_0) = \frac{1}{2} \int_0^T q(x) + u^\top R u \, dt + \Phi(X(T)),$$

where $q(x) > 0$ for all $x \neq 0$ and $q(0) = 0$, and $R = R^\top > 0$. The initial condition of the state is $x(0) = x_0$.

Which of the following statements is/are true?

- ✓ A) The Hamiltonian function associated with this optimal control problem is

$$H(x, u, \lambda) = \frac{1}{2}q(x) + \frac{1}{2}u^\top R u + \lambda^\top (f(x) + g(x)u).$$

- B) The Hamiltonian function associated with this optimal control problem is

$$H(x, u, \lambda) = \frac{1}{2}q(x) + \frac{1}{2}u^\top R u + \lambda^\top (f(x) + g(x)u) + \Phi(x(T)).$$

- ✓ C) The costate λ has dimension n .

- ✓ D) The optimal control is given by

$$u^*(t) = -R^{-1}g(x)^\top \lambda^*(t),$$

where $(x^*(t), \lambda^*(t))$ is the optimal state/costate trajectory given by the solution of

$$\begin{aligned} \dot{x} &= f(x) + g(x)u^* \\ \dot{\lambda}^\top &= -\frac{\partial H}{\partial x}, \end{aligned}$$

subject to $x(0) = x_0$ and $\lambda(T) = \frac{\partial \Phi}{\partial x} \Big|_{x=x(T)}$, where $H(x, u, \lambda)$ denotes the Hamiltonian function associated with the optimal control problem.

Optimal Control – problem formulations

Handout 2 – Solution

1. The correction statements are C) and D). Explanation:

- The term $q(x)$ is part of the running cost that penalises the state trajectory. The requirement that $q(x) > 0$ for $x \neq 0$ and $q(0) = 0$ ensures a penalty is incurred whenever the state is not at the equilibrium (the origin), and incentivises driving the state to zero over time.
- The term $u^\top Ru$ is the part of the running cost that penalises the control effort. The condition $R = R^\top > 0$ (positive definite) ensures that any non-zero control input is penalised. This is necessary to prevent unbounded control effort, making the problem physically meaningful and well-posed.

The two terms detailed above represent balancing the two objectives: driving the state to zero, while minimising the control effort required to do so.

2. The correct options are A) and B)

- A) represents an infinite-horizon discrete-time LQR problem.
- B)) represents finite-horizon discrete-time LQR problem.

Options C) and D) are incorrect, because they involve integrals. Such integrals appear in continuous-time LQR problems, but they are not relevant to the discrete-time setting).

Dynamic Programming

Handout 3 – Solution

1. The correct option is C). The value function is defined as the *optimal cost-to-go*, from some state x_k at time k . The optimal cost-to-go is another way of saying it is the minimum possible cost achievable. Alternatively, it is the cost that would be achieved from state x_k at time k if you were to apply the optimal control as input.
2. The correct answer is A) (see Handout 3, page 40).

Handout 4 – Solution

1. Options A), B) and C) are correct.
 - A) is correct because the value function is time-invariant in the considered setting (the only time-dependence is implicit, via the state $x(t)$).
 - B) follows directly from the definition of value function as the optimal cost-to-go (in this case from x_0 at time $t = 0$).
 - C) is precisely the HJB partial differential equation derived on page 61 of Handout 4, with the only difference being that the vector containing partial derivatives is defined as a *row vector* (hence the transposes in the given equation are different from those in the handout).

Note: Option D) is incorrect, because if the vector containing partial derivatives is defined as a *column* vector, the transposes in the HJB partial differential equation should be different. Namely, the equation reported in D) involves incorrect dimensions. The correct equation with the $\frac{\partial V}{\partial x}$ defined as a column vector would instead be

$$\frac{1}{2}q(x) + \frac{\partial V}{\partial x}^\top f(x) - \frac{1}{2} \frac{\partial V}{\partial x}^\top g(x) R^{-1} g(x)^\top \frac{\partial V}{\partial x} = 0,$$

as reported on page 61 of Handout 4.

LQR

Handout 5 – Solution

1. The correct options is B). This is the difference Riccati equation that arises in the context of finite-horizon, discrete-time LQR problems. Note:
 - Option A is the discrete-time ARE, which arises in *infinite*-horizon discrete-time LQR problems.
 - Option C is the continuous-time ARE, which arises in *infinite*-horizon continuous-time LQR problems.
 - Option D is the differential Riccati equation , which arises in *finite*-horizon continuous-time LQR problems.

Note also that the information given in the preamble is crucial. Since for the stated problem the matrices Q , Q_N and R are all positive-definite, by definition of value function it follows that P must be positive definite.

2. All options are correct (see Sections 4.1 and 4.2 of Handout 5).

Data-driven Control

Handout 6 – Solution

1. The correct option is C). Note that the dimension of the data matrix $\begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix}$ is $(n+m) \times T$ and note that the rank of any matrix cannot exceed the $\hat{n} = \min(\hat{c}, \hat{r})$, where $\hat{c}$ and $\hat{r}$ denotes the number of columns and rows of the matrix. Thus, in this setting the rank cannot exceed $\hat{n} = \min(n+m, T)$. The condition that the data matrix must satisfy is that its

$$\text{rank}\left[\begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix}\right] = n + m$$

(note that this rank cannot exceed $n + m$, irrespective of T). For this to be possible, $T \geq n + m$.

2. The correct option is C). This essentially stems from noting that we can write

$$A + BK = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} I \\ K \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} X_{0,T-1} \\ U_{0,T-1} \end{bmatrix} G_K.$$

The other options are incorrect as they do not capture all the requirements on G_K .

Pontryagin's Minimum Principle

Handout 7 – Solution

1. The correct options are A), C) and D).
 - A) is the definition of the Hamiltonian function (see, e.g. Handout 7, page 101).
 - B) The costate has the same dimension as the state.
 - C) This is precisely what we derived via the calculus of variations (see, e.g. Handout 7, page 107).